QUARTERLY EXAMINATION - 2025 | Exam No.

Time : 3.00 Hours

XII - MATHS

Marks : 90

PART - I

Note: *i) Answer all the questions.*

ii) Choose the best answer and write option code with corresponding answer. **(20x1=20)**

1. If $A = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix}$ then $|\text{adj } (AB)| =$

   1) –40
   2) –80
   3) –60
   4) –20

2. The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ -1 & -2 & -3 & -4 \end{bmatrix}$ is

   1) 1
   2) 2
   3) 4
   4) 3

3. If $A = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$, then $9I_2 - A =$

   1) A^{-1}
   2) $\frac{A^{-1}}{2}$
   3) $3A^{-1}$
   4) $2A^{-1}$

4. If $|Z - 2 + i| \leq 2$, then the greatest value of $|Z|$ is

   1) $\sqrt{3} - 2$
   2) $\sqrt{3} + 2$
   3) $\sqrt{5} - 2$
   4) $\sqrt{5} + 2$

5. The product of all four values of $\left(\cos\frac{\pi}{3} + i \sin\frac{\pi}{3}\right)^{\frac{3}{4}}$ is

   1) –2
   2) –1
   3) 1
   4) 2

6. $\text{Im}(Z\,\bar{Z}) =$

   1) 1
   2) 0
   3) x^2
   4) 3

7. The polynomial $x^3 - kx^2 + 9x$ has three real roots if and only if, k satisfies

   1) $|k| \leq 6$
   2) $k = 0$
   3) $|k| > 6$
   4) $|k| \geq 6$

8. The number of positive roots of the polynomial $\sum_{r=0}^{n} {}^{n}C_r (-1)^r x^r$ is

   1) 0
   2) n
   3) <n
   4) r

9. If $\sin(\sin^{-1} x) = x$

   1) $|x| \leq \frac{\pi}{2}$
   2) $|x| \leq 1$
   3) $|x| > 1$
   4) $|x| < 1$

www.Padasalai.Net

Scanned with Cam

10. $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$ is equal to

1) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ 2) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$

3) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ 4) $\tan^{-1}\left(\frac{1}{2}\right)$

11. If $\sin^{-1} x + \cot^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$, then x is equal to

1) $\frac{1}{2}$ 2) $\frac{1}{\sqrt{5}}$

3) $\frac{2}{\sqrt{5}}$ 4) $\frac{\sqrt{3}}{2}$

12. y=mx+c is a tangent to be the ellipse $\frac{x^2}{a^2} + \frac{x^2}{b^2} = 1$ then

1) $c^2 = a^2m^2 + b^2$ 2) $b^2 = a^2m^2 + c^2$

3) $c^2 = a^2m^2 + m^2$ 4) $c^2 = a^2m^2 - b^2$

13. The radius of the circle passing through the point (6, 2) two of whose diameters are x+y=6 and x+2y=4 is

1) 10 2) $2\sqrt{5}$

3) 6 4) 4

14. An ellipse has OB as semi minor axes, F and F' its foci and the angle FBF' is a right angle. Then the eccentricity of the ellipse is

1) $\frac{1}{\sqrt{2}}$ 2) $\frac{1}{2}$

3) $\frac{1}{4}$ 4) $\frac{1}{\sqrt{3}}$

15. If x+y=k is a normal to the parabola $y^2=12x$, then the value of k is

1) 3 2) −1

3) 1 4) 9

16. The shortest distance between $\vec{r} + \vec{a} = s\vec{u}$ and $\vec{r} + \vec{b} = t\vec{u}$ is

1) $\frac{|(\vec{b} - \vec{a}) \times \vec{u}|}{|\vec{u}|}$ 2) $\frac{(\vec{b} - \vec{a}) \times \vec{u}}{|\vec{u}|}$

3) $\frac{(\vec{b} - \vec{a}) \times \vec{u}}{\vec{u}}$ 4) $\frac{|(\vec{b} - \vec{a}) + \vec{u}|}{|\vec{u}|}$

17. If $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors such that $\vec{a}$ is perpendicular to $\vec{b}$, and is parallel to $\vec{c}$ then $\vec{a} \times (\vec{b} \times \vec{c})$ is equal to

1) $\vec{a}$ 2) $\vec{b}$

3) $\vec{c}$ 4) $\vec{0}$

18. The angle between the lines $\frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ and $\frac{x-1}{1} = \frac{2y+3}{3} = \frac{z+5}{2}$ is

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$

3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

www.Padasalai.Net

Scanned with Cam

19. If the direction cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$, then

1) $c = \pm 3$
2) $c = \pm \sqrt{3}$
3) $c > 0$
4) $0 < c < 1$

20. If $A = \begin{bmatrix} 8 & -4 \\ -5 & 3 \end{bmatrix}$, A (adj A) =

1) 1
2) 2
3) 3
4) 4

PART - II

Note: Answer any seven questions. Question No.30 is compulsory. ***(7x2=14)***

21. Find the rank of the following matrices by minor method $\begin{bmatrix} 1 & -2 & -1 & 0 \\ 3 & -6 & -3 & 1 \end{bmatrix}$.
22. Show that $(2 + i\sqrt{3})^{10} + (2 - i\sqrt{3})^{10}$ is real.
23. Find the square roots of $-6 + 8i$.
24. If α, β, γ are the roots of the equation $x^3+px^2+qx+r=0$, find the value of $\Sigma \frac{1}{\beta\gamma}$ in terms of the coefficients.
25. Find the value of $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$
26. Find the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$.
27. Obtain the equation of the circle for which (3, 4) and (2, –7) are the ends of a diameter.
28. For any vector $\bar{a}$, prove that $\hat{i}\times(\bar{a}\times\hat{i}) + \hat{j}\times(\bar{a}\times\hat{j}) + \hat{k}\times(\bar{a}\times\hat{k}) = 2\bar{a}$.
29. Find the distance of a point (2, 5, –3) from the plane $\bar{r}.(6\hat{i} - 3\hat{j} + 2\hat{k}) = 5$.
30. Find the value of $\frac{(1+i)^{2011}}{(1-i)^{2009}}$.

PART - III

Note: Answer any seven questions. Question No.40 is compulsory. ***(7x3=21)***

31. Solve the following systems of linear equations by Cramer's rule:
$5x-2y+16=0,\ x+3y-7=0$
32. Verify $(AB)^{-1}=B^{-1}A^{-1}$ with $A = \begin{bmatrix} 0 & -3 \\ 1 & 4 \end{bmatrix}, B = \begin{bmatrix} -2 & -3 \\ 0 & -1 \end{bmatrix}$.
33. Find the values of the real numbers x and y, if the complex numbers $(3-i)x-(2-i)y+2i+5$ and $2x+(-1+2i)y+3+2i$ are equal.
34. If $\frac{1+z}{1-z} = \cos 2\theta + i \sin 2\theta$, show that $z = i \tan\theta$.
35. A 12 metre tall tree was broken into two parts. It was found that the height of the part which was left standing was the cube root of the length of the part that was cut away. Formulate this into a mathematical problem to find the height of the part which was left standing.
36. Prove that $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \frac{\pi}{4}$.
37. Find the value of $\cot^{-1}(1) + \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) - \sec^{-1}(-\sqrt{2})$.

www.Padasalai.Net
Scanned with Cam

38. Find the centre and radius of the circle $3x^2+(a+1)y^2+6x-9y+a+4=0$.

39. Prove that the length of the latus rectum of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $\frac{2b^2}{a}$.

40. If $[\bar{a},\bar{b},\bar{c}]=6$, find the value of $[\bar{a}x\bar{b},\bar{b}x\bar{c},\bar{c}x\bar{a}]$.

PART - IV

Note. Answer all the questions. ***(7x5=35)***

41. a) Solve the following system of equations, using matrix inversion method:
$2x_1+3x_2+3x_3=5$, $x_1-2x_2+x_3=-4$, $3x_1-x_2-2x_3=3$

(OR)

b) Solve the equation $z^3+27=0$.

42. a) Find the value of k for which the equations
$kx-2y+z=1$, $x-2ky+z=-2$, $x-2y+kz=1$ have
i) no solution ii) unique solution iii) infinitely many solution

(OR)

b) If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$ and $0<x,y,z<1$, then show that $x^2+y^2+z^2+2xyz=1$.

43. a) Solve the equation $6x^4-5x^3-38x^2-5x+6=0$ if it is known that $\frac{1}{3}$ is a solution.

(OR)

b) If $z=x+iy$ is a complex number such that $\text{Im}\left(\frac{2z+1}{iz+1}\right)=0$, show that $2x^2+2y^2+x-2y=0$.

44. a) Solve the equation $(x-2)(x-7)(x-3)(x+2)+19=0$.

(OR)

b) Prove that $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right]$.

45. a) Assume that water issuing from the end of a horizontal pipe, 7.5m above the ground, describes a parabolic path. The vertex of the parabolic path is at the end of the pipe. At a position 2.5m below the line of the pipe, the flow of water has curved outward 3m beyond the vertical line through the end of the pipe. How far beyond this vertical line will be water strike the ground?

(OR)

b) By vector method, prove that $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$.

46. a) Find the parametric form of vector equation, and Cartesian equations of the plane passing through the points (2, 2, 1), (9, 3, 6) and perpendicular to the plane $2x+6y+6z=9$.

(OR)

b) Show that the line $x-y+4=0$ is a tangent to the ellipse $x^2+3y^2=12$. Also find the coordinate of the point of contact.

47. a) Show that the lines $\frac{x-3}{3}=\frac{y-3}{-1}, z-1=0$ and $\frac{x-6}{2}=\frac{z-1}{3}, y-2=0$ intersect. Also find the point of intersection.

(OR)

b) Find the equation of the circle through the points (1, 0), (–1, 0) and (0, 1).

www.Padasalai.Net

Scanned with Cam